TERTIARY INFOTECH ACADEMY

LAB 12
Capstone — Resolve a Multi-Service CloudOps Incident

AWS Certified SysOps Administrator Associate Training
TGS-2024051413 · Version v5.0

Scenario
After a routine change, Northstar Retail experiences elevated web latency, intermittent 5xx responses, rejected VPC traffic, an incomplete backup job and a security finding.

Objective
Apply A1–A7 in one connected incident by detecting impact, isolating the failed boundary, restoring service, validating data and documenting service-quality improvements.

Deliverable
A timeline, evidence matrix, prioritised hypotheses, recovery runbook, validation record and post-incident action register.

Services and tools
CloudWatch · CloudTrail · VPC Flow Logs · ELB · AWS Backup · Security Hub · Systems Manager

Safety and cost boundary
- Use only the trainer-authorised learner account and Region.
- Replace placeholders and tag learner-owned resources.
- Capture evidence before cleanup.
- Verify deletion and report retained chargeable resources.

Detailed procedure

Step 1 — Read every file in `data/`; record evidence source, timestamp, timezone and reliability in `EVIDENCE-MATRIX.csv`.
```
for f in data/*; do echo "--- $f"; sed -n '1,80p' "$f"; done
```

Evidence note: _____

Step 2 — Run `build_timeline.py`; verify the merged timeline is chronologically sorted and retains source IDs.
```
python3 build_timeline.py | tee evidence/timeline-build.txt
```

Evidence note: _____

Step 3 — Frame impact, scope and declared severity without naming a root cause yet.
```
open INCIDENT-RECORD.md
```

Evidence note: _____

Step 4 — Rank at least three hypotheses and name one discriminating test for each.
```
open HYPOTHESES.csv
```

Evidence note: _____

Step 5 — Identify the first failed boundary by correlating ALB health, Flow Logs, change events and application latency.

```
open output/timeline.csv
```

Evidence note: _____

Step 6 — Review backup and security evidence; decide which items block recovery and which become follow-up actions.

```
open EVIDENCE-MATRIX.csv
```

Evidence note: _____

Step 7 — Complete `RECOVERY-RUNBOOK.md` with authority, exact action, safety check, verification, communication and rollback.

```
open RECOVERY-RUNBOOK.md
```

Evidence note: _____

Step 8 — Execute the trainer-approved simulation action or tabletop step; record UTC timestamps and observed outputs.

```
open RECOVERY-VALIDATION.md
```

Evidence note: _____

Step 9 — Validate service health, request success, data integrity, security control and recovery objective.

```
open RECOVERY-VALIDATION.md
```

Evidence note: _____

Step 10 — Complete `ACTIONS.csv` with preventive, detective and recovery controls, owner, due date and verification threshold.

```
open ACTIONS.csv
```

Evidence note: _____

Step 11 — Deliver a five-minute operational briefing: impact, evidence, decision, recovery, limitation and next controls.

```
open INCIDENT-RECORD.md
```

Evidence note: _____

Step 12 — Capture the final dashboard/architecture evidence as JPEG and run the capstone validator.

```
open EXPECTED-ARCHITECTURE.jpg
bash validate.sh
```

Evidence note: _____

Verification

All evidence IDs reconcile, the timeline precedes causal claims, recovery and rollback are explicit, validation covers service and data, A1–A7 are mapped, and validation emits LAB 12 PASS.

Acceptance criteria

- A1–A7 traceability complete
- Observation and hypothesis separated
- First failed boundary supported
- Recovery includes rollback and communication
- Every action has owner, due date and measurable verification

Evidence checklist

□ timeline.csv

□ evidence matrix

□ hypothesis register

□ incident record

□ recovery runbook

□ recovery validation

□ ACTIONS.csv

□ five-minute briefing

□ JPEG evidence

Troubleshooting

- Check identity, Region and resource ID before changing permissions.
- Resolve every REPLACE placeholder.
- Preserve error output as evidence.
- Escalate cleanup failures immediately.